PVT CORNER TIMING ANALYSIS REPORT

VSDBabySoC Project | Roll No: 24EG104C36

1. Objective

The main aim of this experiment is to check the timing performance of the VSDBabySoC design under different Process, Voltage, and Temperature (PVT) conditions using OpenSTA.

2. Overview

Static Timing Analysis (STA) is used to check whether signals in a digital circuit reach their destination within the required time.

In this experiment, the design was tested at 16 different PVT corners. The timing results were compared using minimum slack, maximum slack, WNS, TNS, and the overall timing status.

This helps identify the operating conditions where the design works properly and where timing problems occur.

3. PVT Corner Comparison Table

S.No	PVT Corner	Min Slack (ns)	Max Slack (ns)	WNS (ns)	TNS (ns)	Status
1	FF_100C_1v65	0.249	4.055	0.000	0.000	MET
2	FF_100C_1v95	0.196	5.567	0.000	0.000	MET
3	FF_n40C_1v56	0.292	2.278	0.000	0.000	MET
4	FF_n40C_1v65	0.255	3.411	0.000	0.000	MET
5	FF_n40C_1v76	0.224	4.430	0.000	0.000	MET
6	FF_n40C_1v95	0.187	5.592	0.000	0.000	MET
7	SS_100C_1v40	0.905	-17.122	-17.122	-2599.514	VIOLATED
8	SS_100C_1v60	0.642	-7.874	-7.874	-892.705	VIOLATED
9	SS_n40C_1v28	1.830	-62.563	-62.563	-13957.588	VIOLATED
10	SS_n40C_1v35	1.348	-39.697	-39.697	-7993.793	VIOLATED
11	SS_n40C_1v40	1.125	-29.694	-29.694	-5466.484	VIOLATED
12	SS_n40C_1v44	0.991	-23.908	-23.908	-4107.187	VIOLATED
13	SS_n40C_1v60	0.663	-10.609	-10.609	-1355.797	VIOLATED
14	SS_n40C_1v76	0.504	-4.382	-4.382	-367.928	VIOLATED
15	TT_025C_1v80	0.310	1.939	0.000	0.000	MET
16	TT_100C_1v80	0.314	2.094	0.000	0.000	MET

4. Timing Graphs

The following graphs show the variation of WNS, TNS, setup slack, and hold slack across the sixteen PVT corners. Numerical values are displayed at each data point for clear comparison.

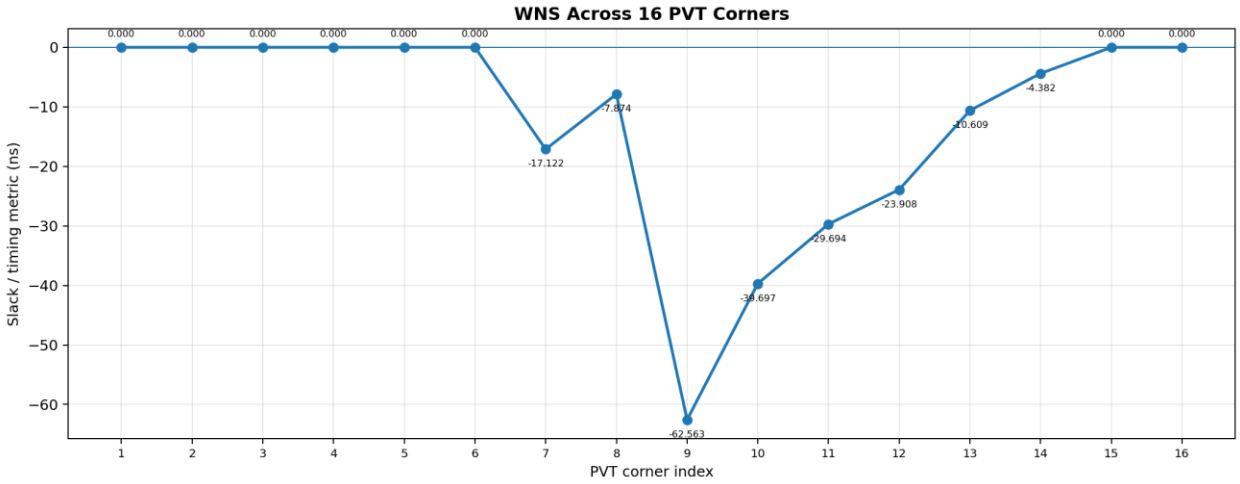


Figure 1: Worst Negative Slack (WNS) across PVT corners

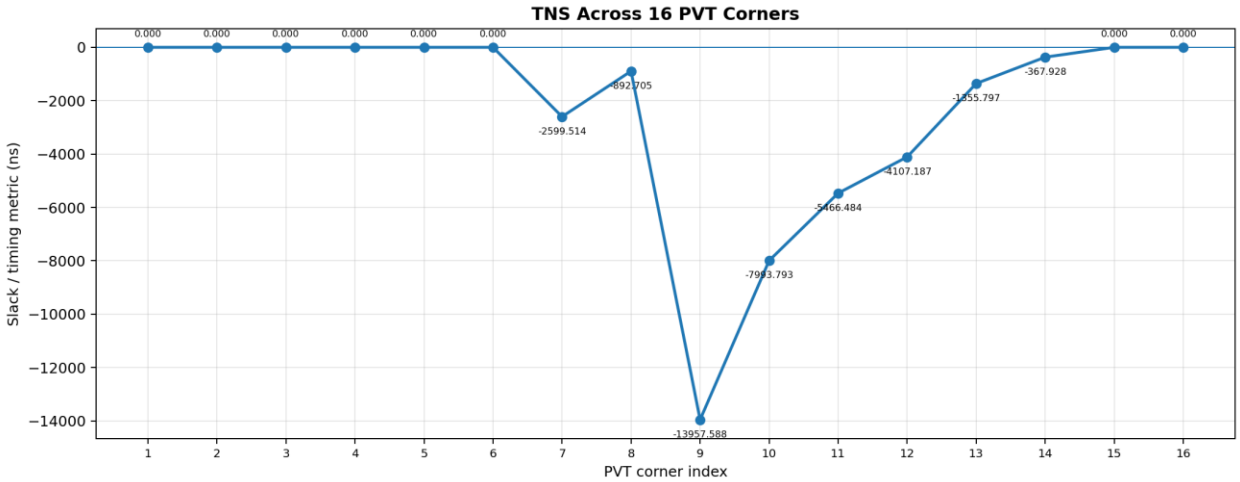


Figure 2: Total Negative Slack (TNS) across PVT corners

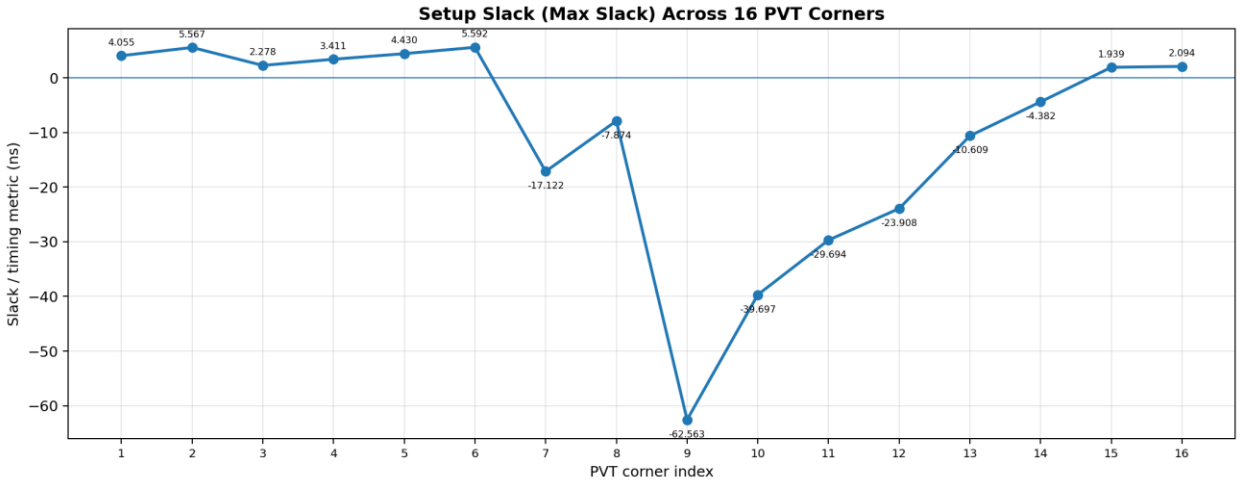


Figure 3: Setup Slack across PVT corners

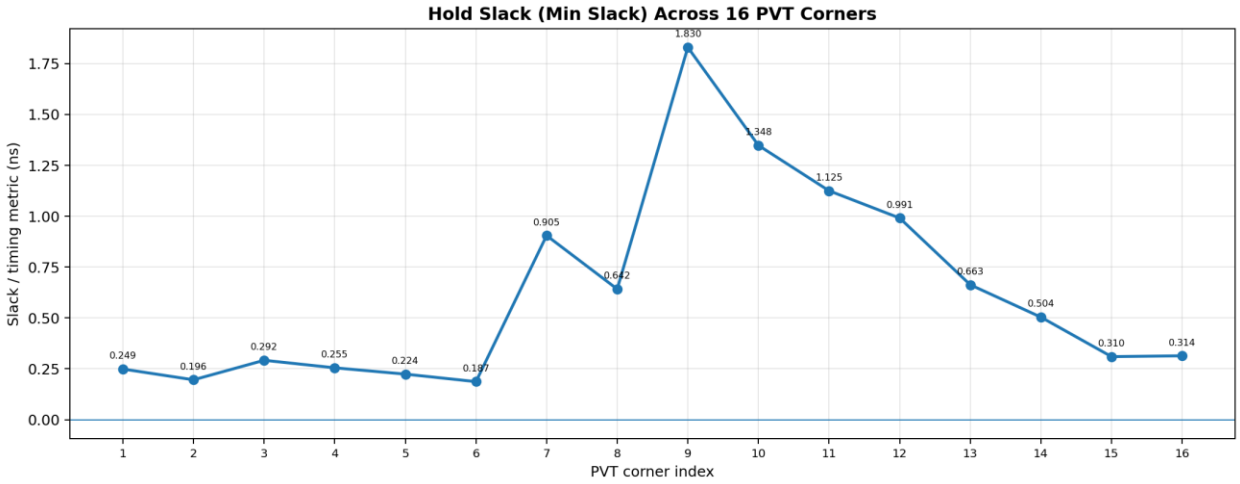


Figure 4: Hold Slack across PVT corners

5. STA REPORTS

Delay Time Description

0.00 0.00 clock clk (rise edge)
0.00 0.00 clock network delay (ideal)
0.00 0.00 ^ core/CPU_is_addi_a3_reg/CLK (sky130_fd_sc_hd__dfxtp_1)
0.24 0.24 v core/CPU_is_addi_a3_reg/Q (sky130_fd_sc_hd__dfxtp_1)
0.09 0.34 ^ core/U474/Y (sky130_fd_sc_hd__nor2_1)
0.08 0.42 v core/U476/Y (sky130_fd_sc_hd__nand2_1)
0.42 0.83 ^ core/U9/Y (sky130_fd_sc_hd__inv_2)
0.13 0.96 v core/U479/X (sky130_fd_sc_hd__xor2_1)
0.19 1.16 ^ core/U569/Y (sky130_fd_sc_hd__a21oi_1)
0.09 1.25 v core/U571/Y (sky130_fd_sc_hd__o21ai_1)
0.15 1.40 ^ core/U574/Y (sky130_fd_sc_hd__a21oi_1)
0.09 1.49 v core/U576/Y (sky130_fd_sc_hd__o21ai_1)
0.15 1.64 ^ core/U581/Y (sky130_fd_sc_hd__a21oi_1)
0.09 1.73 v core/U583/Y (sky130_fd_sc_hd__o21ai_1)
0.15 1.89 ^ core/U587/Y (sky130_fd_sc_hd__a21oi_1)
0.09 1.98 v core/U589/Y (sky130_fd_sc_hd__o21ai_1)
0.15 2.13 ^ core/U344/Y (sky130_fd_sc_hd__a21oi_1)
0.09 2.22 v core/U213/Y (sky130_fd_sc_hd__o21ai_1)
0.15 2.37 ^ core/U343/Y (sky130_fd_sc_hd__a21oi_1)
0.09 2.46 v core/U210/Y (sky130_fd_sc_hd__o21ai_1)
0.15 2.61 ^ core/U342/Y (sky130_fd_sc_hd__a21oi_1)
0.09 2.71 v core/U209/Y (sky130_fd_sc_hd__o21ai_1)
0.15 2.86 ^ core/U341/Y (sky130_fd_sc_hd__a21oi_1)
0.09 2.95 v core/U215/Y (sky130_fd_sc_hd__o21ai_1)
0.14 3.09 v core/U96/X (sky130_fd_sc_hd__a21o_1)
0.28 3.36 v core/U942/COUT (sky130_fd_sc_hd__fa_1)
0.14 3.50 v core/U94/X (sky130_fd_sc_hd__a21o_1)
0.26 3.76 v core/U905/COUT (sky130_fd_sc_hd__fa_1)
0.27 4.03 v core/U887/COUT (sky130_fd_sc_hd__fa_1)
0.27 4.31 v core/U1370/COUT (sky130_fd_sc_hd__fa_1)
0.28 4.59 v core/U36/COUT (sky130_fd_sc_hd__fa_2)
0.04 4.62 ^ core/U87/Y (sky130_fd_sc_hd__clkinv_1)
0.05 4.68 v core/U613/Y (sky130_fd_sc_hd__o21ai_1)

0.26 4.94 v core/U352/COUT (sky130_fd_sc_hd__fa_1)
0.27 5.21 v core/U348/COUT (sky130_fd_sc_hd__fa_1)
0.27 5.48 v core/U35/COUT (sky130_fd_sc_hd__fa_1)
0.29 5.77 v core/U33/COUT (sky130_fd_sc_hd__fa_1)
0.14 5.91 v core/U81/X (sky130_fd_sc_hd__a21o_1)
0.26 6.17 v core/U347/COUT (sky130_fd_sc_hd__fa_1)
0.38 6.55 ^ core/U351/SUM (sky130_fd_sc_hd__fa_1)
0.20 6.75 v core/U3/Y (sky130_fd_sc_hd__nand2_1)
0.13 6.88 ^ core/U696/Y (sky130_fd_sc_hd__o22ai_1)
0.00 6.88 ^ core/CPU_Xreg_value_a4_reg[16][30]/D (sky130_fd_sc_hd__dfxtp_1)
6.88 data arrival time

10.93 data required time

-6.88 data arrival time

4.05 slack (MET)

Startpoint: core/CPU_is_addi_a3_reg
(rising edge-triggered flip-flop clocked by clk)
Endpoint: core/CPU_Xreg_value_a4_reg[16][30]
(rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max
Corner: FF_100C_1v95

Delay Time Description

0.00 0.00 clock clk (rise edge)
0.00 0.00 clock network delay (ideal)
0.00 0.00 ^ core/CPU_is_addi_a3_reg/CLK (sky130_fd_sc_hd__dfxtp_1)
0.20 0.20 v core/CPU_is_addi_a3_reg/Q (sky130_fd_sc_hd__dfxtp_1)
0.07 0.27 ^ core/U474/Y (sky130_fd_sc_hd__nor2_1)
0.07 0.33 v core/U476/Y (sky130_fd_sc_hd__nand2_1)
0.35 0.68 ^ core/U9/Y (sky130_fd_sc_hd__inv_2)
0.09 0.77 v core/U479/X (sky130_fd_sc_hd__xor2_1)
0.15 0.92 ^ core/U569/Y (sky130_fd_sc_hd__a21oi_1)
0.07 0.99 v core/U571/Y (sky130_fd_sc_hd__o21ai_1)
0.12 1.11 ^ core/U574/Y (sky130_fd_sc_hd__a21oi_1)
0.07 1.18 v core/U576/Y (sky130_fd_sc_hd__o21ai_1)
0.12 1.30 ^ core/U581/Y (sky130_fd_sc_hd__a21oi_1)
0.07 1.36 v core/U583/Y (sky130_fd_sc_hd__o21ai_1)
0.12 1.49 ^ core/U587/Y (sky130_fd_sc_hd__a21oi_1)
0.07 1.55 v core/U589/Y (sky130_fd_sc_hd__o21ai_1)
0.12 1.67 ^ core/U344/Y (sky130_fd_sc_hd__a21oi_1)
0.07 1.74 v core/U213/Y (sky130_fd_sc_hd__o21ai_1)
0.12 1.86 ^ core/U343/Y (sky130_fd_sc_hd__a21oi_1)
0.07 1.93 v core/U210/Y (sky130_fd_sc_hd__o21ai_1)
0.12 2.05 ^ core/U342/Y (sky130_fd_sc_hd__a21oi_1)
0.07 2.12 v core/U209/Y (sky130_fd_sc_hd__o21ai_1)
0.12 2.24 ^ core/U341/Y (sky130_fd_sc_hd__a21oi_1)
0.07 2.31 v core/U215/Y (sky130_fd_sc_hd__o21ai_1)
0.11 2.42 v core/U96/X (sky130_fd_sc_hd__a21o_1)
0.22 2.63 v core/U942/COUT (sky130_fd_sc_hd__fa_1)
0.11 2.74 v core/U94/X (sky130_fd_sc_hd__a21o_1)
0.20 2.95 v core/U905/COUT (sky130_fd_sc_hd__fa_1)
0.21 3.16 v core/U887/COUT (sky130_fd_sc_hd__fa_1)
0.21 3.37 v core/U1370/COUT (sky130_fd_sc_hd__fa_1)
0.22 3.58 v core/U36/COUT (sky130_fd_sc_hd__fa_2)

0.03 3.61 ^ core/U87/Y (sky130_fd_sc_hd__clkinv_1)
0.04 3.66 v core/U613/Y (sky130_fd_sc_hd__o21ai_1)
0.21 3.86 v core/U352/COUT (sky130_fd_sc_hd__fa_1)
0.21 4.07 v core/U348/COUT (sky130_fd_sc_hd__fa_1)
0.21 4.28 v core/U35/COUT (sky130_fd_sc_hd__fa_1)
0.22 4.51 v core/U33/COUT (sky130_fd_sc_hd__fa_1)
0.11 4.62 v core/U81/X (sky130_fd_sc_hd__a21o_1)
0.20 4.82 v core/U347/COUT (sky130_fd_sc_hd__fa_1)
0.30 5.12 ^ core/U351/SUM (sky130_fd_sc_hd__fa_1)
0.17 5.28 v core/U3/Y (sky130_fd_sc_hd__nand2_1)
0.11 5.39 ^ core/U696/Y (sky130_fd_sc_hd__o22ai_1)
0.00 5.39 ^ core/CPU_Xreg_value_a4_reg[16][30]/D (sky130_fd_sc_hd__dfxtp_1)
5.39 data arrival time

10.96 data required time

-5.39 data arrival time

5.57 slack (MET)

Startpoint: core/CPU_is_addi_a3_reg
(rising edge-triggered flip-flop clocked by clk)
Endpoint: core/CPU_Xreg_value_a4_reg[16][31]
(rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max
Corner: FF_n40C_1v56

Delay Time Description

0.00 0.00 clock clk (rise edge)
0.00 0.00 clock network delay (ideal)
0.00 0.00 ^ core/CPU_is_addi_a3_reg/CLK (sky130_fd_sc_hd__dfxtp_1)
0.27 0.27 v core/CPU_is_addi_a3_reg/Q (sky130_fd_sc_hd__dfxtp_1)
0.13 0.39 ^ core/U474/Y (sky130_fd_sc_hd__nor2_1)
0.08 0.48 v core/U476/Y (sky130_fd_sc_hd__nand2_1)
0.54 1.01 ^ core/U9/Y (sky130_fd_sc_hd__inv_2)
0.15 1.16 v core/U479/X (sky130_fd_sc_hd__xor2_1)
0.27 1.43 ^ core/U569/Y (sky130_fd_sc_hd__a21oi_1)
0.10 1.53 v core/U571/Y (sky130_fd_sc_hd__o21ai_1)
0.21 1.74 ^ core/U574/Y (sky130_fd_sc_hd__a21oi_1)
0.10 1.84 v core/U576/Y (sky130_fd_sc_hd__o21ai_1)
0.21 2.04 ^ core/U581/Y (sky130_fd_sc_hd__a21oi_1)
0.10 2.14 v core/U583/Y (sky130_fd_sc_hd__o21ai_1)
0.21 2.35 ^ core/U587/Y (sky130_fd_sc_hd__a21oi_1)
0.10 2.45 v core/U589/Y (sky130_fd_sc_hd__o21ai_1)
0.21 2.65 ^ core/U344/Y (sky130_fd_sc_hd__a21oi_1)
0.10 2.75 v core/U213/Y (sky130_fd_sc_hd__o21ai_1)
0.21 2.96 ^ core/U343/Y (sky130_fd_sc_hd__a21oi_1)
0.10 3.06 v core/U210/Y (sky130_fd_sc_hd__o21ai_1)
0.21 3.26 ^ core/U342/Y (sky130_fd_sc_hd__a21oi_1)
0.10 3.36 v core/U209/Y (sky130_fd_sc_hd__o21ai_1)
0.21 3.57 ^ core/U341/Y (sky130_fd_sc_hd__a21oi_1)
0.10 3.67 v core/U215/Y (sky130_fd_sc_hd__o21ai_1)
0.17 3.84 v core/U96/X (sky130_fd_sc_hd__a21o_1)
0.35 4.19 v core/U942/COUT (sky130_fd_sc_hd__fa_1)
0.18 4.38 v core/U94/X (sky130_fd_sc_hd__a21o_1)
0.33 4.71 v core/U905/COUT (sky130_fd_sc_hd__fa_1)
0.35 5.06 v core/U887/COUT (sky130_fd_sc_hd__fa_1)

0.35 5.41 v core/U1370/COUT (sky130_fd_sc_hd__fa_1)
0.34 5.74 v core/U36/COUT (sky130_fd_sc_hd__fa_2)
0.06 5.80 ^ core/U87/Y (sky130_fd_sc_hd__clkinv_1)
0.05 5.85 v core/U613/Y (sky130_fd_sc_hd__o21ai_1)
0.34 6.18 v core/U352/COUT (sky130_fd_sc_hd__fa_1)
0.35 6.53 v core/U348/COUT (sky130_fd_sc_hd__fa_1)
0.35 6.88 v core/U35/COUT (sky130_fd_sc_hd__fa_1)
0.36 7.24 v core/U33/COUT (sky130_fd_sc_hd__fa_1)
0.18 7.42 v core/U81/X (sky130_fd_sc_hd__a21o_1)
0.33 7.76 v core/U347/COUT (sky130_fd_sc_hd__fa_1)
0.34 8.10 v core/U351/COUT (sky130_fd_sc_hd__fa_1)
0.22 8.32 ^ core/U617/X (sky130_fd_sc_hd__xor2_1)
0.16 8.48 v core/U618/Y (sky130_fd_sc_hd__nand2_2)
0.16 8.64 ^ core/U635/Y (sky130_fd_sc_hd__o22ai_1)
0.00 8.64 ^ core/CPU_Xreg_value_a4_reg[16][31]/D (sky130_fd_sc_hd__dfxtp_1)
8.64 data arrival time

10.92 data required time

-8.64 data arrival time

2.28 slack (MET)

Startpoint: core/CPU_is_addi_a3_reg
(rising edge-triggered flip-flop clocked by clk)
Endpoint: core/CPU_Xreg_value_a4_reg[16][31]
(rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max
Corner: FF_n40C_1v65

Delay Time Description

0.00 0.00 clock clk (rise edge)
0.00 0.00 clock network delay (ideal)
0.00 0.00 ^ core/CPU_is_addi_a3_reg/CLK (sky130_fd_sc_hd__dfxtp_1)
0.24 0.24 v core/CPU_is_addi_a3_reg/Q (sky130_fd_sc_hd__dfxtp_1)
0.11 0.35 ^ core/U474/Y (sky130_fd_sc_hd__nor2_1)
0.07 0.42 v core/U476/Y (sky130_fd_sc_hd__nand2_1)
0.48 0.90 ^ core/U9/Y (sky130_fd_sc_hd__inv_2)
0.12 1.02 v core/U479/X (sky130_fd_sc_hd__xor2_1)
0.23 1.26 ^ core/U569/Y (sky130_fd_sc_hd__a21oi_1)
0.09 1.34 v core/U571/Y (sky130_fd_sc_hd__o21ai_1)
0.18 1.53 ^ core/U574/Y (sky130_fd_sc_hd__a21oi_1)
0.09 1.61 v core/U576/Y (sky130_fd_sc_hd__o21ai_1)
0.18 1.79 ^ core/U581/Y (sky130_fd_sc_hd__a21oi_1)
0.09 1.88 v core/U583/Y (sky130_fd_sc_hd__o21ai_1)
0.18 2.06 ^ core/U587/Y (sky130_fd_sc_hd__a21oi_1)
0.09 2.15 v core/U589/Y (sky130_fd_sc_hd__o21ai_1)
0.18 2.33 ^ core/U344/Y (sky130_fd_sc_hd__a21oi_1)
0.09 2.41 v core/U213/Y (sky130_fd_sc_hd__o21ai_1)
0.18 2.59 ^ core/U343/Y (sky130_fd_sc_hd__a21oi_1)
0.09 2.68 v core/U210/Y (sky130_fd_sc_hd__o21ai_1)
0.18 2.86 ^ core/U342/Y (sky130_fd_sc_hd__a21oi_1)
0.09 2.95 v core/U209/Y (sky130_fd_sc_hd__o21ai_1)
0.18 3.13 ^ core/U341/Y (sky130_fd_sc_hd__a21oi_1)
0.09 3.22 v core/U215/Y (sky130_fd_sc_hd__o21ai_1)
0.15 3.37 v core/U96/X (sky130_fd_sc_hd__a21o_1)
0.30 3.67 v core/U942/COUT (sky130_fd_sc_hd__fa_1)

0.16 3.83 v core/U94/X (sky130_fd_sc_hd__a21o_1)
0.29 4.12 v core/U905/COUT (sky130_fd_sc_hd__fa_1)
0.30 4.42 v core/U887/COUT (sky130_fd_sc_hd__fa_1)
0.30 4.72 v core/U1370/COUT (sky130_fd_sc_hd__fa_1)
0.29 5.01 v core/U36/COUT (sky130_fd_sc_hd__fa_2)
0.05 5.06 ^ core/U87/Y (sky130_fd_sc_hd__clkinv_1)
0.04 5.10 v core/U613/Y (sky130_fd_sc_hd__o21ai_1)
0.29 5.39 v core/U352/COUT (sky130_fd_sc_hd__fa_1)
0.30 5.69 v core/U348/COUT (sky130_fd_sc_hd__fa_1)
0.30 5.99 v core/U35/COUT (sky130_fd_sc_hd__fa_1)
0.31 6.30 v core/U33/COUT (sky130_fd_sc_hd__fa_1)
0.16 6.46 v core/U81/X (sky130_fd_sc_hd__a21o_1)
0.29 6.75 v core/U347/COUT (sky130_fd_sc_hd__fa_1)
0.30 7.05 v core/U351/COUT (sky130_fd_sc_hd__fa_1)
0.19 7.24 ^ core/U617/X (sky130_fd_sc_hd__xor2_1)
0.14 7.38 v core/U618/Y (sky130_fd_sc_hd__nand2_2)
0.14 7.52 ^ core/U635/Y (sky130_fd_sc_hd__o22ai_1)
0.00 7.52 ^ core/CPU_Xreg_value_a4_reg[16][31]/D (sky130_fd_sc_hd__dfxtp_1)
7.52 data arrival time

10.93 data required time

-7.52 data arrival time

3.41 slack (MET)

Startpoint: core/CPU_is_addi_a3_reg
(rising edge-triggered flip-flop clocked by clk)
Endpoint: core/CPU_Xreg_value_a4_reg[1][31]
(rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max
Corner: FF_n40C_1v76

Delay Time Description

0.00 0.00 clock clk (rise edge)
0.00 0.00 clock network delay (ideal)
0.00 0.00 ^ core/CPU_is_addi_a3_reg/CLK (sky130_fd_sc_hd__dfxtp_1)
0.21 0.21 v core/CPU_is_addi_a3_reg/Q (sky130_fd_sc_hd__dfxtp_1)
0.10 0.31 ^ core/U474/Y (sky130_fd_sc_hd__nor2_1)
0.06 0.37 v core/U476/Y (sky130_fd_sc_hd__nand2_1)
0.43 0.81 ^ core/U9/Y (sky130_fd_sc_hd__inv_2)
0.09 0.90 v core/U479/X (sky130_fd_sc_hd__xor2_1)
0.20 1.10 ^ core/U569/Y (sky130_fd_sc_hd__a21oi_1)
0.07 1.17 v core/U571/Y (sky130_fd_sc_hd__o21ai_1)
0.16 1.33 ^ core/U574/Y (sky130_fd_sc_hd__a21oi_1)
0.07 1.41 v core/U576/Y (sky130_fd_sc_hd__o21ai_1)
0.16 1.56 ^ core/U581/Y (sky130_fd_sc_hd__a21oi_1)
0.07 1.64 v core/U583/Y (sky130_fd_sc_hd__o21ai_1)
0.16 1.80 ^ core/U587/Y (sky130_fd_sc_hd__a21oi_1)
0.07 1.87 v core/U589/Y (sky130_fd_sc_hd__o21ai_1)
0.16 2.03 ^ core/U344/Y (sky130_fd_sc_hd__a21oi_1)
0.07 2.10 v core/U213/Y (sky130_fd_sc_hd__o21ai_1)
0.16 2.26 ^ core/U343/Y (sky130_fd_sc_hd__a21oi_1)
0.07 2.33 v core/U210/Y (sky130_fd_sc_hd__o21ai_1)
0.16 2.49 ^ core/U342/Y (sky130_fd_sc_hd__a21oi_1)
0.07 2.56 v core/U209/Y (sky130_fd_sc_hd__o21ai_1)
0.16 2.72 ^ core/U341/Y (sky130_fd_sc_hd__a21oi_1)

0.07 2.80 v core/U215/Y (sky130_fd_sc_hd__o21ai_1)
0.13 2.93 v core/U96/X (sky130_fd_sc_hd__a21o_1)
0.26 3.19 v core/U942/COUT (sky130_fd_sc_hd__fa_1)
0.14 3.33 v core/U94/X (sky130_fd_sc_hd__a21o_1)
0.25 3.58 v core/U905/COUT (sky130_fd_sc_hd__fa_1)
0.26 3.83 v core/U887/COUT (sky130_fd_sc_hd__fa_1)
0.26 4.09 v core/U1370/COUT (sky130_fd_sc_hd__fa_1)
0.25 4.34 v core/U36/COUT (sky130_fd_sc_hd__fa_2)
0.04 4.38 ^ core/U87/Y (sky130_fd_sc_hd__clkinv_1)
0.04 4.42 v core/U613/Y (sky130_fd_sc_hd__o21ai_1)
0.25 4.68 v core/U352/COUT (sky130_fd_sc_hd__fa_1)
0.26 4.93 v core/U348/COUT (sky130_fd_sc_hd__fa_1)
0.26 5.19 v core/U35/COUT (sky130_fd_sc_hd__fa_1)
0.27 5.46 v core/U33/COUT (sky130_fd_sc_hd__fa_1)
0.14 5.60 v core/U81/X (sky130_fd_sc_hd__a21o_1)
0.25 5.85 v core/U347/COUT (sky130_fd_sc_hd__fa_1)
0.25 6.10 v core/U351/COUT (sky130_fd_sc_hd__fa_1)
0.17 6.27 ^ core/U617/X (sky130_fd_sc_hd__xor2_1)
0.12 6.39 v core/U618/Y (sky130_fd_sc_hd__nand2_2)
0.12 6.52 ^ core/U643/Y (sky130_fd_sc_hd__o22ai_1)
0.00 6.52 ^ core/CPU_Xreg_value_a4_reg[1][31]/D (sky130_fd_sc_hd__dfxtp_1)
6.52 data arrival time

10.95 data required time

-6.52 data arrival time

4.43 slack (MET)

Startpoint: core/CPU_is_addi_a3_reg
(rising edge-triggered flip-flop clocked by clk)
Endpoint: core/CPU_Xreg_value_a4_reg[16][31]
(rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max
Corner: FF_n40C_1v95

Delay Time Description

0.00 0.00 clock clk (rise edge)
0.00 0.00 clock network delay (ideal)
0.00 0.00 ^ core/CPU_is_addi_a3_reg/CLK (sky130_fd_sc_hd__dfxtp_1)
0.18 0.18 v core/CPU_is_addi_a3_reg/Q (sky130_fd_sc_hd__dfxtp_1)
0.08 0.26 ^ core/U474/Y (sky130_fd_sc_hd__nor2_1)
0.05 0.32 v core/U476/Y (sky130_fd_sc_hd__nand2_1)
0.37 0.69 ^ core/U9/Y (sky130_fd_sc_hd__inv_2)
0.06 0.75 v core/U479/X (sky130_fd_sc_hd__xor2_1)
0.17 0.92 ^ core/U569/Y (sky130_fd_sc_hd__a21oi_1)
0.06 0.98 v core/U571/Y (sky130_fd_sc_hd__o21ai_1)
0.13 1.11 ^ core/U574/Y (sky130_fd_sc_hd__a21oi_1)
0.06 1.17 v core/U576/Y (sky130_fd_sc_hd__o21ai_1)
0.13 1.30 ^ core/U581/Y (sky130_fd_sc_hd__a21oi_1)
0.06 1.36 v core/U583/Y (sky130_fd_sc_hd__o21ai_1)
0.13 1.49 ^ core/U587/Y (sky130_fd_sc_hd__a21oi_1)
0.06 1.55 v core/U589/Y (sky130_fd_sc_hd__o21ai_1)
0.13 1.68 ^ core/U344/Y (sky130_fd_sc_hd__a21oi_1)
0.06 1.74 v core/U213/Y (sky130_fd_sc_hd__o21ai_1)
0.13 1.87 ^ core/U343/Y (sky130_fd_sc_hd__a21oi_1)
0.06 1.93 v core/U210/Y (sky130_fd_sc_hd__o21ai_1)

0.13 2.06 ^ core/U342/Y (sky130_fd_sc_hd__a21oi_1)
0.06 2.12 v core/U209/Y (sky130_fd_sc_hd__o21ai_1)
0.13 2.26 ^ core/U341/Y (sky130_fd_sc_hd__a21oi_1)
0.06 2.32 v core/U215/Y (sky130_fd_sc_hd__o21ai_1)
0.11 2.43 v core/U96/X (sky130_fd_sc_hd__a21o_1)
0.21 2.64 v core/U942/COUT (sky130_fd_sc_hd__fa_1)
0.11 2.75 v core/U94/X (sky130_fd_sc_hd__a21o_1)
0.20 2.96 v core/U905/COUT (sky130_fd_sc_hd__fa_1)
0.21 3.17 v core/U887/COUT (sky130_fd_sc_hd__fa_1)
0.21 3.38 v core/U1370/COUT (sky130_fd_sc_hd__fa_1)
0.20 3.58 v core/U36/COUT (sky130_fd_sc_hd__fa_2)
0.03 3.62 ^ core/U87/Y (sky130_fd_sc_hd__clkinv_1)
0.04 3.65 v core/U613/Y (sky130_fd_sc_hd__o21ai_1)
0.21 3.86 v core/U352/COUT (sky130_fd_sc_hd__fa_1)
0.21 4.07 v core/U348/COUT (sky130_fd_sc_hd__fa_1)
0.21 4.28 v core/U35/COUT (sky130_fd_sc_hd__fa_1)
0.22 4.50 v core/U33/COUT (sky130_fd_sc_hd__fa_1)
0.11 4.61 v core/U81/X (sky130_fd_sc_hd__a21o_1)
0.20 4.82 v core/U347/COUT (sky130_fd_sc_hd__fa_1)
0.21 5.02 v core/U351/COUT (sky130_fd_sc_hd__fa_1)
0.14 5.17 ^ core/U617/X (sky130_fd_sc_hd__xor2_1)
0.10 5.27 v core/U618/Y (sky130_fd_sc_hd__nand2_2)
0.10 5.37 ^ core/U635/Y (sky130_fd_sc_hd__o22ai_1)
0.00 5.37 ^ core/CPU_Xreg_value_a4_reg[16][31]/D (sky130_fd_sc_hd__dfxtp_1)
5.37 data arrival time

10.96 data required time

-5.37 data arrival time

5.59 slack (MET)

Startpoint: core/CPU_is_addi_a3_reg
(rising edge-triggered flip-flop clocked by clk)
Endpoint: core/CPU_Xreg_value_a4_reg[27][30]
(rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max
Corner: SS_100C_1v40

Delay Time Description

0.00 0.00 clock clk (rise edge)
0.00 0.00 clock network delay (ideal)
0.00 0.00 ^ core/CPU_is_addi_a3_reg/CLK (sky130_fd_sc_hd__dfxtp_1)
0.91 0.91 ^ core/CPU_is_addi_a3_reg/Q (sky130_fd_sc_hd__dfxtp_1)
0.21 1.13 v core/U54/Y (sky130_fd_sc_hd__clkinv_1)
0.18 1.30 ^ core/U477/Y (sky130_fd_sc_hd__nand2_1)
0.71 2.01 v core/U51/Y (sky130_fd_sc_hd__nand2_2)
0.80 2.81 v core/U478/X (sky130_fd_sc_hd__a22o_1)
0.47 3.28 v core/U479/X (sky130_fd_sc_hd__xor2_1)
0.58 3.86 ^ core/U569/Y (sky130_fd_sc_hd__a21oi_1)
0.36 4.22 v core/U571/Y (sky130_fd_sc_hd__o21ai_1)
0.48 4.70 ^ core/U574/Y (sky130_fd_sc_hd__a21oi_1)
0.36 5.05 v core/U576/Y (sky130_fd_sc_hd__o21ai_1)
0.48 5.53 ^ core/U581/Y (sky130_fd_sc_hd__a21oi_1)
0.36 5.89 v core/U583/Y (sky130_fd_sc_hd__o21ai_1)
0.48 6.37 ^ core/U587/Y (sky130_fd_sc_hd__a21oi_1)
0.36 6.72 v core/U589/Y (sky130_fd_sc_hd__o21ai_1)

0.48 7.20 ^ core/U344/Y (sky130_fd_sc_hd__a21oi_1)
0.36 7.56 v core/U213/Y (sky130_fd_sc_hd__o21ai_1)
0.48 8.04 ^ core/U343/Y (sky130_fd_sc_hd__a21oi_1)
0.36 8.39 v core/U210/Y (sky130_fd_sc_hd__o21ai_1)
0.48 8.87 ^ core/U342/Y (sky130_fd_sc_hd__a21oi_1)
0.36 9.23 v core/U209/Y (sky130_fd_sc_hd__o21ai_1)
0.48 9.71 ^ core/U341/Y (sky130_fd_sc_hd__a21oi_1)
0.36 10.06 v core/U215/Y (sky130_fd_sc_hd__o21ai_1)
0.49 10.55 v core/U96/X (sky130_fd_sc_hd__a21o_1)
1.32 11.87 v core/U942/COUT (sky130_fd_sc_hd__fa_1)
0.51 12.38 v core/U94/X (sky130_fd_sc_hd__a21o_1)
1.27 13.65 v core/U905/COUT (sky130_fd_sc_hd__fa_1)
1.33 14.98 v core/U887/COUT (sky130_fd_sc_hd__fa_1)
1.34 16.32 v core/U1370/COUT (sky130_fd_sc_hd__fa_1)
1.13 17.45 v core/U36/COUT (sky130_fd_sc_hd__fa_2)
0.13 17.58 ^ core/U87/Y (sky130_fd_sc_hd__clkinv_1)
0.17 17.75 v core/U613/Y (sky130_fd_sc_hd__o21ai_1)
1.30 19.05 v core/U352/COUT (sky130_fd_sc_hd__fa_1)
1.33 20.38 v core/U348/COUT (sky130_fd_sc_hd__fa_1)
1.33 21.70 v core/U35/COUT (sky130_fd_sc_hd__fa_1)
1.37 23.08 v core/U33/COUT (sky130_fd_sc_hd__fa_1)
0.51 23.59 v core/U81/X (sky130_fd_sc_hd__a21o_1)
1.27 24.86 v core/U347/COUT (sky130_fd_sc_hd__fa_1)
1.63 26.49 v core/U351/SUM (sky130_fd_sc_hd__fa_1)
0.66 27.15 ^ core/U3/Y (sky130_fd_sc_hd__nand2_1)
0.39 27.54 v core/U684/Y (sky130_fd_sc_hd__o22ai_1)
0.00 27.54 v core/CPU_Xreg_value_a4_reg[27][30]/D (sky130_fd_sc_hd__dfxtp_1)
27.54 data arrival time

10.42 data required time

-27.54 data arrival time

-17.12 slack (VIOLATED)

Startpoint: core/CPU_is_addi_a3_reg
(rising edge-triggered flip-flop clocked by clk)
Endpoint: core/CPU_Xreg_value_a4_reg[27][30]
(rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max
Corner: SS_100C_1v60

Delay Time Description

0.00 0.00 clock clk (rise edge)
0.00 0.00 clock network delay (ideal)
0.00 0.00 ^ core/CPU_is_addi_a3_reg/CLK (sky130_fd_sc_hd__dfxtp_1)
0.62 0.62 ^ core/CPU_is_addi_a3_reg/Q (sky130_fd_sc_hd__dfxtp_1)
0.15 0.78 v core/U54/Y (sky130_fd_sc_hd__clkinv_1)
0.13 0.91 ^ core/U477/Y (sky130_fd_sc_hd__nand2_1)
0.51 1.42 v core/U51/Y (sky130_fd_sc_hd__nand2_2)
0.56 1.98 v core/U478/X (sky130_fd_sc_hd__a22o_1)
0.33 2.31 v core/U479/X (sky130_fd_sc_hd__xor2_1)
0.42 2.72 ^ core/U569/Y (sky130_fd_sc_hd__a21oi_1)
0.24 2.97 v core/U571/Y (sky130_fd_sc_hd__o21ai_1)
0.34 3.31 ^ core/U574/Y (sky130_fd_sc_hd__a21oi_1)
0.24 3.55 v core/U576/Y (sky130_fd_sc_hd__o21ai_1)
0.34 3.89 ^ core/U581/Y (sky130_fd_sc_hd__a21oi_1)

0.24 4.13 v core/U583/Y (sky130_fd_sc_hd__o21ai_1)
0.34 4.47 ^ core/U587/Y (sky130_fd_sc_hd__a21oi_1)
0.24 4.72 v core/U589/Y (sky130_fd_sc_hd__o21ai_1)
0.34 5.06 ^ core/U344/Y (sky130_fd_sc_hd__a21oi_1)
0.24 5.30 v core/U213/Y (sky130_fd_sc_hd__o21ai_1)
0.34 5.64 ^ core/U343/Y (sky130_fd_sc_hd__a21oi_1)
0.24 5.88 v core/U210/Y (sky130_fd_sc_hd__o21ai_1)
0.34 6.22 ^ core/U342/Y (sky130_fd_sc_hd__a21oi_1)
0.24 6.46 v core/U209/Y (sky130_fd_sc_hd__o21ai_1)
0.34 6.81 ^ core/U341/Y (sky130_fd_sc_hd__a21oi_1)
0.24 7.05 v core/U215/Y (sky130_fd_sc_hd__o21ai_1)
0.34 7.38 v core/U96/X (sky130_fd_sc_hd__a21o_1)
0.86 8.24 v core/U942/COUT (sky130_fd_sc_hd__fa_1)
0.35 8.59 v core/U94/X (sky130_fd_sc_hd__a21o_1)
0.83 9.42 v core/U905/COUT (sky130_fd_sc_hd__fa_1)
0.86 10.28 v core/U887/COUT (sky130_fd_sc_hd__fa_1)
0.87 11.14 v core/U1370/COUT (sky130_fd_sc_hd__fa_1)
0.74 11.89 v core/U36/COUT (sky130_fd_sc_hd__fa_2)
0.09 11.98 ^ core/U87/Y (sky130_fd_sc_hd__clkinv_1)
0.12 12.10 v core/U613/Y (sky130_fd_sc_hd__o21ai_1)
0.84 12.94 v core/U352/COUT (sky130_fd_sc_hd__fa_1)
0.86 13.80 v core/U348/COUT (sky130_fd_sc_hd__fa_1)
0.86 14.66 v core/U35/COUT (sky130_fd_sc_hd__fa_1)
0.89 15.55 v core/U33/COUT (sky130_fd_sc_hd__fa_1)
0.35 15.90 v core/U81/X (sky130_fd_sc_hd__a21o_1)
0.83 16.73 v core/U347/COUT (sky130_fd_sc_hd__fa_1)
1.04 17.77 v core/U351/SUM (sky130_fd_sc_hd__fa_1)
0.48 18.25 ^ core/U3/Y (sky130_fd_sc_hd__nand2_1)
0.26 18.51 v core/U684/Y (sky130_fd_sc_hd__o22ai_1)
0.00 18.51 v core/CPU_Xreg_value_a4_reg[27][30]/D (sky130_fd_sc_hd__dfxtp_1)
18.51 data arrival time

10.64 data required time

-18.51 data arrival time

-7.87 slack (VIOLATED)

Startpoint: core/CPU_is_addi_a3_reg
(rising edge-triggered flip-flop clocked by clk)
Endpoint: core/CPU_Xreg_value_a4_reg[27][30]
(rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max
Corner: SS_n40C_1v28

Delay Time Description

0.00 0.00 clock clk (rise edge)
0.00 0.00 clock network delay (ideal)
0.00 0.00 ^ core/CPU_is_addi_a3_reg/CLK (sky130_fd_sc_hd__dfxtp_1)
1.50 1.50 v core/CPU_is_addi_a3_reg/Q (sky130_fd_sc_hd__dfxtp_1)
0.71 2.20 ^ core/U474/Y (sky130_fd_sc_hd__nor2_1)
0.56 2.76 v core/U476/Y (sky130_fd_sc_hd__nand2_1)
2.67 5.43 ^ core/U9/Y (sky130_fd_sc_hd__inv_2)
1.71 7.14 v core/U479/X (sky130_fd_sc_hd__xor2_1)
1.42 8.57 ^ core/U569/Y (sky130_fd_sc_hd__a21oi_1)
0.76 9.33 v core/U571/Y (sky130_fd_sc_hd__o21ai_1)
1.20 10.53 ^ core/U574/Y (sky130_fd_sc_hd__a21oi_1)

0.76 11.29 v core/U576/Y (sky130_fd_sc_hd__o21ai_1)
1.20 12.49 ^ core/U581/Y (sky130_fd_sc_hd__a21oi_1)
0.76 13.25 v core/U583/Y (sky130_fd_sc_hd__o21ai_1)
1.20 14.45 ^ core/U587/Y (sky130_fd_sc_hd__a21oi_1)
0.76 15.21 v core/U589/Y (sky130_fd_sc_hd__o21ai_1)
1.20 16.41 ^ core/U344/Y (sky130_fd_sc_hd__a21oi_1)
0.76 17.17 v core/U213/Y (sky130_fd_sc_hd__o21ai_1)
1.20 18.38 ^ core/U343/Y (sky130_fd_sc_hd__a21oi_1)
0.76 19.14 v core/U210/Y (sky130_fd_sc_hd__o21ai_1)
1.20 20.34 ^ core/U342/Y (sky130_fd_sc_hd__a21oi_1)
0.76 21.10 v core/U209/Y (sky130_fd_sc_hd__o21ai_1)
1.20 22.30 ^ core/U341/Y (sky130_fd_sc_hd__a21oi_1)
0.76 23.06 v core/U215/Y (sky130_fd_sc_hd__o21ai_1)
1.13 24.19 v core/U96/X (sky130_fd_sc_hd__a21o_1)
3.70 27.89 v core/U942/COUT (sky130_fd_sc_hd__fa_1)
1.32 29.21 v core/U94/X (sky130_fd_sc_hd__a21o_1)
3.62 32.83 v core/U905/COUT (sky130_fd_sc_hd__fa_1)
3.78 36.61 v core/U887/COUT (sky130_fd_sc_hd__fa_1)
3.80 40.42 v core/U1370/COUT (sky130_fd_sc_hd__fa_1)
3.39 43.81 v core/U36/COUT (sky130_fd_sc_hd__fa_2)
0.40 44.21 ^ core/U87/Y (sky130_fd_sc_hd__clkinv_1)
0.21 44.43 v core/U613/Y (sky130_fd_sc_hd__o21ai_1)
3.63 48.06 v core/U352/COUT (sky130_fd_sc_hd__fa_1)
3.78 51.84 v core/U348/COUT (sky130_fd_sc_hd__fa_1)
3.78 55.62 v core/U35/COUT (sky130_fd_sc_hd__fa_1)
3.86 59.48 v core/U33/COUT (sky130_fd_sc_hd__fa_1)
1.32 60.80 v core/U81/X (sky130_fd_sc_hd__a21o_1)
3.62 64.42 v core/U347/COUT (sky130_fd_sc_hd__fa_1)
4.76 69.19 v core/U351/SUM (sky130_fd_sc_hd__fa_1)
1.61 70.80 ^ core/U3/Y (sky130_fd_sc_hd__nand2_1)
0.91 71.70 v core/U684/Y (sky130_fd_sc_hd__o22ai_1)
0.00 71.70 v core/CPU_Xreg_value_a4_reg[27][30]/D (sky130_fd_sc_hd__dfxtp_1)
71.70 data arrival time

9.14 data required time

-71.70 data arrival time

-62.56 slack (VIOLATED)

Startpoint: core/CPU_is_addi_a3_reg
(rising edge-triggered flip-flop clocked by clk)
Endpoint: core/CPU_Xreg_value_a4_reg[27][30]
(rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max
Corner: SS_n40C_1v35

Delay Time Description

0.00 0.00 clock clk (rise edge)
0.00 0.00 clock network delay (ideal)
0.00 0.00 ^ core/CPU_is_addi_a3_reg/CLK (sky130_fd_sc_hd__dfxtp_1)
1.12 1.12 v core/CPU_is_addi_a3_reg/Q (sky130_fd_sc_hd__dfxtp_1)
0.52 1.64 ^ core/U474/Y (sky130_fd_sc_hd__nor2_1)
0.41 2.05 v core/U476/Y (sky130_fd_sc_hd__nand2_1)
2.01 4.06 ^ core/U9/Y (sky130_fd_sc_hd__inv_2)
1.22 5.28 v core/U479/X (sky130_fd_sc_hd__xor2_1)
1.04 6.33 ^ core/U569/Y (sky130_fd_sc_hd__a21oi_1)

0.54 6.87 v core/U571/Y (sky130_fd_sc_hd__o21ai_1)
0.87 7.74 ^ core/U574/Y (sky130_fd_sc_hd__a21oi_1)
0.54 8.28 v core/U576/Y (sky130_fd_sc_hd__o21ai_1)
0.87 9.15 ^ core/U581/Y (sky130_fd_sc_hd__a21oi_1)
0.54 9.69 v core/U583/Y (sky130_fd_sc_hd__o21ai_1)
0.87 10.56 ^ core/U587/Y (sky130_fd_sc_hd__a21oi_1)
0.54 11.10 v core/U589/Y (sky130_fd_sc_hd__o21ai_1)
0.87 11.96 ^ core/U344/Y (sky130_fd_sc_hd__a21oi_1)
0.54 12.51 v core/U213/Y (sky130_fd_sc_hd__o21ai_1)
0.87 13.37 ^ core/U343/Y (sky130_fd_sc_hd__a21oi_1)
0.54 13.92 v core/U210/Y (sky130_fd_sc_hd__o21ai_1)
0.87 14.78 ^ core/U342/Y (sky130_fd_sc_hd__a21oi_1)
0.54 15.33 v core/U209/Y (sky130_fd_sc_hd__o21ai_1)
0.87 16.19 ^ core/U341/Y (sky130_fd_sc_hd__a21oi_1)
0.54 16.74 v core/U215/Y (sky130_fd_sc_hd__o21ai_1)
0.81 17.54 v core/U96/X (sky130_fd_sc_hd__a21o_1)
2.47 20.01 v core/U942/COUT (sky130_fd_sc_hd__fa_1)
0.91 20.92 v core/U94/X (sky130_fd_sc_hd__a21o_1)
2.42 23.34 v core/U905/COUT (sky130_fd_sc_hd__fa_1)
2.52 25.87 v core/U887/COUT (sky130_fd_sc_hd__fa_1)
2.54 28.40 v core/U1370/COUT (sky130_fd_sc_hd__fa_1)
2.27 30.67 v core/U36/COUT (sky130_fd_sc_hd__fa_2)
0.27 30.94 ^ core/U87/Y (sky130_fd_sc_hd__clkinv_1)
0.17 31.11 v core/U613/Y (sky130_fd_sc_hd__o21ai_1)
2.43 33.54 v core/U352/COUT (sky130_fd_sc_hd__fa_1)
2.52 36.07 v core/U348/COUT (sky130_fd_sc_hd__fa_1)
2.52 38.59 v core/U35/COUT (sky130_fd_sc_hd__fa_1)
2.57 41.16 v core/U33/COUT (sky130_fd_sc_hd__fa_1)
0.91 42.07 v core/U81/X (sky130_fd_sc_hd__a21o_1)
2.42 44.49 v core/U347/COUT (sky130_fd_sc_hd__fa_1)
3.14 47.63 v core/U351/SUM (sky130_fd_sc_hd__fa_1)
1.20 48.83 ^ core/U3/Y (sky130_fd_sc_hd__nand2_1)
0.64 49.47 v core/U684/Y (sky130_fd_sc_hd__o22ai_1)
0.00 49.47 v core/CPU_Xreg_value_a4_reg[27][30]/D (sky130_fd_sc_hd__dfxtp_1)
49.47 data arrival time

9.77 data required time

-49.47 data arrival time

-39.70 slack (VIOLATED)

Startpoint: core/CPU_is_addi_a3_reg
(rising edge-triggered flip-flop clocked by clk)
Endpoint: core/CPU_Xreg_value_a4_reg[27][30]
(rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max
Corner: SS_n40C_1v40

Delay Time Description

0.00 0.00 clock clk (rise edge)
0.00 0.00 clock network delay (ideal)
0.00 0.00 ^ core/CPU_is_addi_a3_reg/CLK (sky130_fd_sc_hd__dfxtp_1)
0.94 0.94 v core/CPU_is_addi_a3_reg/Q (sky130_fd_sc_hd__dfxtp_1)
0.43 1.38 ^ core/U474/Y (sky130_fd_sc_hd__nor2_1)
0.34 1.72 v core/U476/Y (sky130_fd_sc_hd__nand2_1)
1.70 3.42 ^ core/U9/Y (sky130_fd_sc_hd__inv_2)

0.99 4.40 v core/U479/X (sky130_fd_sc_hd__xor2_1)
0.87 5.27 ^ core/U569/Y (sky130_fd_sc_hd__a21oi_1)
0.44 5.71 v core/U571/Y (sky130_fd_sc_hd__o21ai_1)
0.71 6.43 ^ core/U574/Y (sky130_fd_sc_hd__a21oi_1)
0.44 6.87 v core/U576/Y (sky130_fd_sc_hd__o21ai_1)
0.71 7.58 ^ core/U581/Y (sky130_fd_sc_hd__a21oi_1)
0.44 8.02 v core/U583/Y (sky130_fd_sc_hd__o21ai_1)
0.71 8.74 ^ core/U587/Y (sky130_fd_sc_hd__a21oi_1)
0.44 9.18 v core/U589/Y (sky130_fd_sc_hd__o21ai_1)
0.71 9.89 ^ core/U344/Y (sky130_fd_sc_hd__a21oi_1)
0.44 10.33 v core/U213/Y (sky130_fd_sc_hd__o21ai_1)
0.71 11.05 ^ core/U343/Y (sky130_fd_sc_hd__a21oi_1)
0.44 11.49 v core/U210/Y (sky130_fd_sc_hd__o21ai_1)
0.71 12.20 ^ core/U342/Y (sky130_fd_sc_hd__a21oi_1)
0.44 12.64 v core/U209/Y (sky130_fd_sc_hd__o21ai_1)
0.71 13.36 ^ core/U341/Y (sky130_fd_sc_hd__a21oi_1)
0.44 13.80 v core/U215/Y (sky130_fd_sc_hd__o21ai_1)
0.66 14.46 v core/U96/X (sky130_fd_sc_hd__a21o_1)
1.95 16.41 v core/U942/COUT (sky130_fd_sc_hd__fa_1)
0.74 17.14 v core/U94/X (sky130_fd_sc_hd__a21o_1)
1.91 19.05 v core/U905/COUT (sky130_fd_sc_hd__fa_1)
1.98 21.04 v core/U887/COUT (sky130_fd_sc_hd__fa_1)
2.00 23.03 v core/U1370/COUT (sky130_fd_sc_hd__fa_1)
1.79 24.82 v core/U36/COUT (sky130_fd_sc_hd__fa_2)
0.22 25.04 ^ core/U87/Y (sky130_fd_sc_hd__clkinv_1)
0.14 25.18 v core/U613/Y (sky130_fd_sc_hd__o21ai_1)
1.92 27.10 v core/U352/COUT (sky130_fd_sc_hd__fa_1)
1.98 29.09 v core/U348/COUT (sky130_fd_sc_hd__fa_1)
1.98 31.07 v core/U35/COUT (sky130_fd_sc_hd__fa_1)
2.02 33.10 v core/U33/COUT (sky130_fd_sc_hd__fa_1)
0.74 33.83 v core/U81/X (sky130_fd_sc_hd__a21o_1)
1.91 35.74 v core/U347/COUT (sky130_fd_sc_hd__fa_1)
2.46 38.20 v core/U351/SUM (sky130_fd_sc_hd__fa_1)
1.00 39.20 ^ core/U3/Y (sky130_fd_sc_hd__nand2_1)
0.53 39.74 v core/U684/Y (sky130_fd_sc_hd__o22ai_1)
0.00 39.74 v core/CPU_Xreg_value_a4_reg[27][30]/D (sky130_fd_sc_hd__dfxtp_1)
39.74 data arrival time

10.04 data required time

-39.74 data arrival time

-29.69 slack (VIOLATED)

Startpoint: core/CPU_is_addi_a3_reg
(rising edge-triggered flip-flop clocked by clk)
Endpoint: core/CPU_Xreg_value_a4_reg[27][30]
(rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max
Corner: SS_n40C_1v44

Delay Time Description

0.00 0.00 clock clk (rise edge)
0.00 0.00 clock network delay (ideal)
0.00 0.00 ^ core/CPU_is_addi_a3_reg/CLK (sky130_fd_sc_hd__dfxtp_1)
0.83 0.83 v core/CPU_is_addi_a3_reg/Q (sky130_fd_sc_hd__dfxtp_1)
0.38 1.21 ^ core/U474/Y (sky130_fd_sc_hd__nor2_1)

0.30 1.51 v core/U476/Y (sky130_fd_sc_hd__nand2_1)
1.50 3.01 ^ core/U9/Y (sky130_fd_sc_hd__inv_2)
0.88 3.88 v core/U479/X (sky130_fd_sc_hd__xor2_1)
0.77 4.65 ^ core/U569/Y (sky130_fd_sc_hd__a21oi_1)
0.39 5.04 v core/U571/Y (sky130_fd_sc_hd__o21ai_1)
0.62 5.67 ^ core/U574/Y (sky130_fd_sc_hd__a21oi_1)
0.39 6.06 v core/U576/Y (sky130_fd_sc_hd__o21ai_1)
0.62 6.68 ^ core/U581/Y (sky130_fd_sc_hd__a21oi_1)
0.39 7.07 v core/U583/Y (sky130_fd_sc_hd__o21ai_1)
0.62 7.69 ^ core/U587/Y (sky130_fd_sc_hd__a21oi_1)
0.39 8.08 v core/U589/Y (sky130_fd_sc_hd__o21ai_1)
0.62 8.71 ^ core/U344/Y (sky130_fd_sc_hd__a21oi_1)
0.39 9.10 v core/U213/Y (sky130_fd_sc_hd__o21ai_1)
0.62 9.72 ^ core/U343/Y (sky130_fd_sc_hd__a21oi_1)
0.39 10.11 v core/U210/Y (sky130_fd_sc_hd__o21ai_1)
0.62 10.73 ^ core/U342/Y (sky130_fd_sc_hd__a21oi_1)
0.39 11.12 v core/U209/Y (sky130_fd_sc_hd__o21ai_1)
0.62 11.75 ^ core/U341/Y (sky130_fd_sc_hd__a21oi_1)
0.39 12.14 v core/U215/Y (sky130_fd_sc_hd__o21ai_1)
0.57 12.71 v core/U96/X (sky130_fd_sc_hd__a21o_1)
1.65 14.36 v core/U942/COUT (sky130_fd_sc_hd__fa_1)
0.63 15.00 v core/U94/X (sky130_fd_sc_hd__a21o_1)
1.61 16.61 v core/U905/COUT (sky130_fd_sc_hd__fa_1)
1.67 18.28 v core/U887/COUT (sky130_fd_sc_hd__fa_1)
1.68 19.96 v core/U1370/COUT (sky130_fd_sc_hd__fa_1)
1.51 21.48 v core/U36/COUT (sky130_fd_sc_hd__fa_2)
0.19 21.67 ^ core/U87/Y (sky130_fd_sc_hd__clkinv_1)
0.13 21.79 v core/U613/Y (sky130_fd_sc_hd__o21ai_1)
1.62 23.42 v core/U352/COUT (sky130_fd_sc_hd__fa_1)
1.67 25.09 v core/U348/COUT (sky130_fd_sc_hd__fa_1)
1.67 26.76 v core/U35/COUT (sky130_fd_sc_hd__fa_1)
1.71 28.47 v core/U33/COUT (sky130_fd_sc_hd__fa_1)
0.63 29.10 v core/U81/X (sky130_fd_sc_hd__a21o_1)
1.61 30.72 v core/U347/COUT (sky130_fd_sc_hd__fa_1)
2.07 32.79 v core/U351/SUM (sky130_fd_sc_hd__fa_1)
0.87 33.65 ^ core/U3/Y (sky130_fd_sc_hd__nand2_1)
0.45 34.10 v core/U684/Y (sky130_fd_sc_hd__o22ai_1)
0.00 34.10 v core/CPU_Xreg_value_a4_reg[27][30]/D (sky130_fd_sc_hd__dfxtp_1)
34.10 data arrival time

10.20 data required time

-34.10 data arrival time

-23.91 slack (VIOLATED)

Startpoint: core/CPU_is_addi_a3_reg
(rising edge-triggered flip-flop clocked by clk)
Endpoint: core/CPU_Xreg_value_a4_reg[27][30]
(rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max
Corner: SS_n40C_1v60

Delay Time Description

0.00 0.00 clock clk (rise edge)
0.00 0.00 clock network delay (ideal)
0.00 0.00 ^ core/CPU_is_addi_a3_reg/CLK (sky130_fd_sc_hd__dfxtp_1)

0.57 0.57 v core/CPU_is_addi_a3_reg/Q (sky130_fd_sc_hd__dfxtp_1)
0.25 0.82 ^ core/U474/Y (sky130_fd_sc_hd__nor2_1)
0.20 1.02 v core/U476/Y (sky130_fd_sc_hd__nand2_1)
1.04 2.06 ^ core/U9/Y (sky130_fd_sc_hd__inv_2)
0.53 2.58 v core/U479/X (sky130_fd_sc_hd__xor2_1)
0.52 3.10 ^ core/U569/Y (sky130_fd_sc_hd__a21oi_1)
0.24 3.35 v core/U571/Y (sky130_fd_sc_hd__o21ai_1)
0.41 3.76 ^ core/U574/Y (sky130_fd_sc_hd__a21oi_1)
0.24 4.00 v core/U576/Y (sky130_fd_sc_hd__o21ai_1)
0.41 4.41 ^ core/U581/Y (sky130_fd_sc_hd__a21oi_1)
0.24 4.66 v core/U583/Y (sky130_fd_sc_hd__o21ai_1)
0.41 5.06 ^ core/U587/Y (sky130_fd_sc_hd__a21oi_1)
0.24 5.31 v core/U589/Y (sky130_fd_sc_hd__o21ai_1)
0.41 5.72 ^ core/U344/Y (sky130_fd_sc_hd__a21oi_1)
0.24 5.96 v core/U213/Y (sky130_fd_sc_hd__o21ai_1)
0.41 6.37 ^ core/U343/Y (sky130_fd_sc_hd__a21oi_1)
0.24 6.62 v core/U210/Y (sky130_fd_sc_hd__o21ai_1)
0.41 7.03 ^ core/U342/Y (sky130_fd_sc_hd__a21oi_1)
0.24 7.27 v core/U209/Y (sky130_fd_sc_hd__o21ai_1)
0.41 7.68 ^ core/U341/Y (sky130_fd_sc_hd__a21oi_1)
0.25 7.93 v core/U215/Y (sky130_fd_sc_hd__o21ai_1)
0.37 8.30 v core/U96/X (sky130_fd_sc_hd__a21o_1)
0.98 9.28 v core/U942/COUT (sky130_fd_sc_hd__fa_1)
0.40 9.68 v core/U94/X (sky130_fd_sc_hd__a21o_1)
0.96 10.64 v core/U905/COUT (sky130_fd_sc_hd__fa_1)
0.99 11.63 v core/U887/COUT (sky130_fd_sc_hd__fa_1)
1.00 12.63 v core/U1370/COUT (sky130_fd_sc_hd__fa_1)
0.91 13.53 v core/U36/COUT (sky130_fd_sc_hd__fa_2)
0.12 13.65 ^ core/U87/Y (sky130_fd_sc_hd__clkinv_1)
0.09 13.74 v core/U613/Y (sky130_fd_sc_hd__o21ai_1)
0.96 14.71 v core/U352/COUT (sky130_fd_sc_hd__fa_1)
0.99 15.70 v core/U348/COUT (sky130_fd_sc_hd__fa_1)
0.99 16.69 v core/U35/COUT (sky130_fd_sc_hd__fa_1)
1.02 17.71 v core/U33/COUT (sky130_fd_sc_hd__fa_1)
0.40 18.11 v core/U81/X (sky130_fd_sc_hd__a21o_1)
0.96 19.06 v core/U347/COUT (sky130_fd_sc_hd__fa_1)
1.21 20.27 v core/U351/SUM (sky130_fd_sc_hd__fa_1)
0.59 20.87 ^ core/U3/Y (sky130_fd_sc_hd__nand2_1)
0.29 21.15 v core/U684/Y (sky130_fd_sc_hd__o22ai_1)
0.00 21.15 v core/CPU_Xreg_value_a4_reg[27][30]/D (sky130_fd_sc_hd__dfxtp_1)
21.15 data arrival time

10.54 data required time

-21.15 data arrival time

-10.61 slack (VIOLATED)

Startpoint: core/CPU_is_addi_a3_reg
(rising edge-triggered flip-flop clocked by clk)
Endpoint: core/CPU_Xreg_value_a4_reg[27][30]
(rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max
Corner: SS_n40C_1v76

Delay Time Description

0.00 0.00 clock clk (rise edge)
0.00 0.00 clock network delay (ideal)
0.00 0.00 ^ core/CPU_is_addi_a3_reg/CLK (sky130_fd_sc_hd__dfxtp_1)
0.43 0.43 v core/CPU_is_addi_a3_reg/Q (sky130_fd_sc_hd__dfxtp_1)
0.19 0.62 ^ core/U474/Y (sky130_fd_sc_hd__nor2_1)
0.15 0.77 v core/U476/Y (sky130_fd_sc_hd__nand2_1)
0.80 1.57 ^ core/U9/Y (sky130_fd_sc_hd__inv_2)
0.36 1.93 v core/U479/X (sky130_fd_sc_hd__xor2_1)
0.39 2.32 ^ core/U569/Y (sky130_fd_sc_hd__a21oi_1)
0.18 2.50 v core/U571/Y (sky130_fd_sc_hd__o21ai_1)
0.31 2.80 ^ core/U574/Y (sky130_fd_sc_hd__a21oi_1)
0.18 2.98 v core/U576/Y (sky130_fd_sc_hd__o21ai_1)
0.31 3.29 ^ core/U581/Y (sky130_fd_sc_hd__a21oi_1)
0.18 3.47 v core/U583/Y (sky130_fd_sc_hd__o21ai_1)
0.31 3.77 ^ core/U587/Y (sky130_fd_sc_hd__a21oi_1)
0.18 3.95 v core/U589/Y (sky130_fd_sc_hd__o21ai_1)
0.31 4.26 ^ core/U344/Y (sky130_fd_sc_hd__a21oi_1)
0.18 4.44 v core/U213/Y (sky130_fd_sc_hd__o21ai_1)
0.31 4.74 ^ core/U343/Y (sky130_fd_sc_hd__a21oi_1)
0.18 4.92 v core/U210/Y (sky130_fd_sc_hd__o21ai_1)
0.31 5.23 ^ core/U342/Y (sky130_fd_sc_hd__a21oi_1)
0.18 5.41 v core/U209/Y (sky130_fd_sc_hd__o21ai_1)
0.31 5.71 ^ core/U341/Y (sky130_fd_sc_hd__a21oi_1)
0.18 5.89 v core/U215/Y (sky130_fd_sc_hd__o21ai_1)
0.27 6.16 v core/U96/X (sky130_fd_sc_hd__a21o_1)
0.68 6.84 v core/U942/COUT (sky130_fd_sc_hd__fa_1)
0.29 7.13 v core/U94/X (sky130_fd_sc_hd__a21o_1)
0.65 7.79 v core/U905/COUT (sky130_fd_sc_hd__fa_1)
0.68 8.47 v core/U887/COUT (sky130_fd_sc_hd__fa_1)
0.68 9.15 v core/U1370/COUT (sky130_fd_sc_hd__fa_1)
0.63 9.78 v core/U36/COUT (sky130_fd_sc_hd__fa_2)
0.09 9.87 ^ core/U87/Y (sky130_fd_sc_hd__clkinv_1)
0.07 9.94 v core/U613/Y (sky130_fd_sc_hd__o21ai_1)
0.66 10.61 v core/U352/COUT (sky130_fd_sc_hd__fa_1)
0.68 11.28 v core/U348/COUT (sky130_fd_sc_hd__fa_1)
0.68 11.96 v core/U35/COUT (sky130_fd_sc_hd__fa_1)
0.70 12.66 v core/U33/COUT (sky130_fd_sc_hd__fa_1)
0.29 12.95 v core/U81/X (sky130_fd_sc_hd__a21o_1)
0.65 13.61 v core/U347/COUT (sky130_fd_sc_hd__fa_1)
0.82 14.43 v core/U351/SUM (sky130_fd_sc_hd__fa_1)
0.45 14.88 ^ core/U3/Y (sky130_fd_sc_hd__nand2_1)
0.19 15.07 v core/U684/Y (sky130_fd_sc_hd__o22ai_1)
0.00 15.07 v core/CPU_Xreg_value_a4_reg[27][30]/D (sky130_fd_sc_hd__dfxtp_1)
15.07 data arrival time

10.69 data required time

-15.07 data arrival time

-4.38 slack (VIOLATED)

Startpoint: core/CPU_is_addi_a3_reg
(rising edge-triggered flip-flop clocked by clk)
Endpoint: core/CPU_Xreg_value_a4_reg[27][30]
(rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max
Corner: TT_025C_1v80

Delay Time Description

0.00 0.00 clock clk (rise edge)
0.00 0.00 clock network delay (ideal)
0.00 0.00 ^ core/CPU_is_addi_a3_reg/CLK (sky130_fd_sc_hd__dfxtp_1)
0.28 0.28 v core/CPU_is_addi_a3_reg/Q (sky130_fd_sc_hd__dfxtp_1)
0.12 0.40 ^ core/U474/Y (sky130_fd_sc_hd__nor2_1)
0.10 0.50 v core/U476/Y (sky130_fd_sc_hd__nand2_1)
0.51 1.01 ^ core/U9/Y (sky130_fd_sc_hd__inv_2)
0.18 1.19 v core/U479/X (sky130_fd_sc_hd__xor2_1)
0.25 1.43 ^ core/U569/Y (sky130_fd_sc_hd__a21oi_1)
0.11 1.54 v core/U571/Y (sky130_fd_sc_hd__o21ai_1)
0.19 1.74 ^ core/U574/Y (sky130_fd_sc_hd__a21oi_1)
0.11 1.84 v core/U576/Y (sky130_fd_sc_hd__o21ai_1)
0.19 2.04 ^ core/U581/Y (sky130_fd_sc_hd__a21oi_1)
0.11 2.15 v core/U583/Y (sky130_fd_sc_hd__o21ai_1)
0.19 2.34 ^ core/U587/Y (sky130_fd_sc_hd__a21oi_1)
0.11 2.45 v core/U589/Y (sky130_fd_sc_hd__o21ai_1)
0.19 2.64 ^ core/U344/Y (sky130_fd_sc_hd__a21oi_1)
0.11 2.75 v core/U213/Y (sky130_fd_sc_hd__o21ai_1)
0.19 2.94 ^ core/U343/Y (sky130_fd_sc_hd__a21oi_1)
0.11 3.05 v core/U210/Y (sky130_fd_sc_hd__o21ai_1)
0.19 3.24 ^ core/U342/Y (sky130_fd_sc_hd__a21oi_1)
0.11 3.35 v core/U209/Y (sky130_fd_sc_hd__o21ai_1)
0.19 3.54 ^ core/U341/Y (sky130_fd_sc_hd__a21oi_1)
0.11 3.65 v core/U215/Y (sky130_fd_sc_hd__o21ai_1)
0.17 3.82 v core/U96/X (sky130_fd_sc_hd__a21o_1)
0.39 4.21 v core/U942/COUT (sky130_fd_sc_hd__fa_1)
0.18 4.39 v core/U94/X (sky130_fd_sc_hd__a21o_1)
0.37 4.75 v core/U905/COUT (sky130_fd_sc_hd__fa_1)
0.38 5.13 v core/U887/COUT (sky130_fd_sc_hd__fa_1)
0.38 5.52 v core/U1370/COUT (sky130_fd_sc_hd__fa_1)
0.36 5.87 v core/U36/COUT (sky130_fd_sc_hd__fa_2)
0.05 5.93 ^ core/U87/Y (sky130_fd_sc_hd__clkinv_1)
0.06 5.98 v core/U613/Y (sky130_fd_sc_hd__o21ai_1)
0.37 6.35 v core/U352/COUT (sky130_fd_sc_hd__fa_1)
0.38 6.73 v core/U348/COUT (sky130_fd_sc_hd__fa_1)
0.38 7.11 v core/U35/COUT (sky130_fd_sc_hd__fa_1)
0.40 7.51 v core/U33/COUT (sky130_fd_sc_hd__fa_1)
0.18 7.69 v core/U81/X (sky130_fd_sc_hd__a21o_1)
0.37 8.06 v core/U347/COUT (sky130_fd_sc_hd__fa_1)
0.45 8.51 v core/U351/SUM (sky130_fd_sc_hd__fa_1)
0.29 8.80 ^ core/U3/Y (sky130_fd_sc_hd__nand2_1)
0.10 8.90 v core/U684/Y (sky130_fd_sc_hd__o22ai_1)
0.00 8.90 v core/CPU_Xreg_value_a4_reg[27][30]/D (sky130_fd_sc_hd__dfxtp_1)
8.90 data arrival time

10.84 data required time

-8.90 data arrival time

1.94 slack (MET)

Startpoint: core/CPU_is_addi_a3_reg
(rising edge-triggered flip-flop clocked by clk)
Endpoint: core/CPU_Xreg_value_a4_reg[27][30]
(rising edge-triggered flip-flop clocked by clk)
Path Group: clk

Path Type: max
Corner: TT_100C_1v80

Delay Time Description

0.00 0.00 clock clk (rise edge)
0.00 0.00 clock network delay (ideal)
0.00 0.00 ^ core/CPU_is_addi_a3_reg/CLK (sky130_fd_sc_hd__dfxtp_1)
0.29 0.29 v core/CPU_is_addi_a3_reg/Q (sky130_fd_sc_hd__dfxtp_1)
0.11 0.40 ^ core/U474/Y (sky130_fd_sc_hd__nor2_1)
0.10 0.51 v core/U476/Y (sky130_fd_sc_hd__nand2_1)
0.48 0.99 ^ core/U9/Y (sky130_fd_sc_hd__inv_2)
0.19 1.18 v core/U479/X (sky130_fd_sc_hd__xor2_1)
0.23 1.41 ^ core/U569/Y (sky130_fd_sc_hd__a21oi_1)
0.12 1.53 v core/U571/Y (sky130_fd_sc_hd__o21ai_1)
0.18 1.71 ^ core/U574/Y (sky130_fd_sc_hd__a21oi_1)
0.11 1.83 v core/U576/Y (sky130_fd_sc_hd__o21ai_1)
0.18 2.01 ^ core/U581/Y (sky130_fd_sc_hd__a21oi_1)
0.11 2.12 v core/U583/Y (sky130_fd_sc_hd__o21ai_1)
0.18 2.31 ^ core/U587/Y (sky130_fd_sc_hd__a21oi_1)
0.11 2.42 v core/U589/Y (sky130_fd_sc_hd__o21ai_1)
0.18 2.61 ^ core/U344/Y (sky130_fd_sc_hd__a21oi_1)
0.11 2.72 v core/U213/Y (sky130_fd_sc_hd__o21ai_1)
0.18 2.90 ^ core/U343/Y (sky130_fd_sc_hd__a21oi_1)
0.11 3.02 v core/U210/Y (sky130_fd_sc_hd__o21ai_1)
0.18 3.20 ^ core/U342/Y (sky130_fd_sc_hd__a21oi_1)
0.11 3.32 v core/U209/Y (sky130_fd_sc_hd__o21ai_1)
0.18 3.50 ^ core/U341/Y (sky130_fd_sc_hd__a21oi_1)
0.11 3.62 v core/U215/Y (sky130_fd_sc_hd__o21ai_1)
0.17 3.79 v core/U96/X (sky130_fd_sc_hd__a21o_1)
0.38 4.16 v core/U942/COUT (sky130_fd_sc_hd__fa_1)
0.17 4.33 v core/U94/X (sky130_fd_sc_hd__a21o_1)
0.36 4.69 v core/U905/COUT (sky130_fd_sc_hd__fa_1)
0.37 5.06 v core/U887/COUT (sky130_fd_sc_hd__fa_1)
0.37 5.43 v core/U1370/COUT (sky130_fd_sc_hd__fa_1)
0.35 5.78 v core/U36/COUT (sky130_fd_sc_hd__fa_2)
0.05 5.83 ^ core/U87/Y (sky130_fd_sc_hd__clkinv_1)
0.06 5.89 v core/U613/Y (sky130_fd_sc_hd__o21ai_1)
0.36 6.26 v core/U352/COUT (sky130_fd_sc_hd__fa_1)
0.37 6.62 v core/U348/COUT (sky130_fd_sc_hd__fa_1)
0.37 6.99 v core/U35/COUT (sky130_fd_sc_hd__fa_1)
0.39 7.38 v core/U33/COUT (sky130_fd_sc_hd__fa_1)
0.17 7.55 v core/U81/X (sky130_fd_sc_hd__a21o_1)
0.36 7.91 v core/U347/COUT (sky130_fd_sc_hd__fa_1)
0.49 8.40 ^ core/U351/SUM (sky130_fd_sc_hd__fa_1)
0.25 8.65 v core/U3/Y (sky130_fd_sc_hd__nand2_1)
0.16 8.81 ^ core/U684/Y (sky130_fd_sc_hd__o22ai_1)
0.00 8.81 ^ core/CPU_Xreg_value_a4_reg[27][30]/D (sky130_fd_sc_hd__dfxtp_1)
8.81 data arrival time

10.90 data required time

-8.81 data arrival time

2.09 slack (MET)

6. Observations

1. All six fast-process (FF) corners passed the reported timing checks.
2. Both typical-process (TT) corners also passed the reported timing checks.
3. All eight slow-process (SS) corners showed negative WNS and TNS values.
4. The largest setup timing violation occurred at SS_n40C_1v28, with a WNS of -62.563 ns.
5. The results show that the circuit experiences greater delay under slow-process and low-voltage conditions.
6. The timing performance changes depending on the process, supply voltage, and temperature.

7. Conclusion

The timing analysis of the VSDBabySoC design was performed successfully using OpenSTA across 16 PVT corners.

The fast and typical corners passed the reported timing checks, while the slow corners showed setup timing violations.

The results show that the design requires further timing optimization under slow operating conditions. Improvements such as reducing critical path delay, optimizing logic cells, and using suitable buffering may help improve timing performance.

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